TMM Lab 06: Practitioner Studies

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Introduction

This lab sheet assumes you have engaged with the material for the practitioner studies topic shared on Blackboard.

Having read through the case studies and attended/replayed the lecture, you should have a sense of the considerations necessary for completing a full life cycle of a transport-related project that involves modelling, including relevant stakeholders, processes, guidance, and possibly commercial tools.

We recommend that you work in groups for this lab sheet. Please use the lab session, and talk to the lecturers to develop answer strategies for each of the questions. Starred questions are more ambitious and should be skipped if you are not making good progress.

The challenge

In the hypothetical scenario we consider here, the University of Bristol and the City Council have started discussions about making the university precinct more pedestrian friendly. In addition to pedestrianizing Tyndall avenue, a smaller intervention, namely the possibility of a pedestrian crossing between the Merchant Venturers Building and Queens Building, is being considered (see map below).

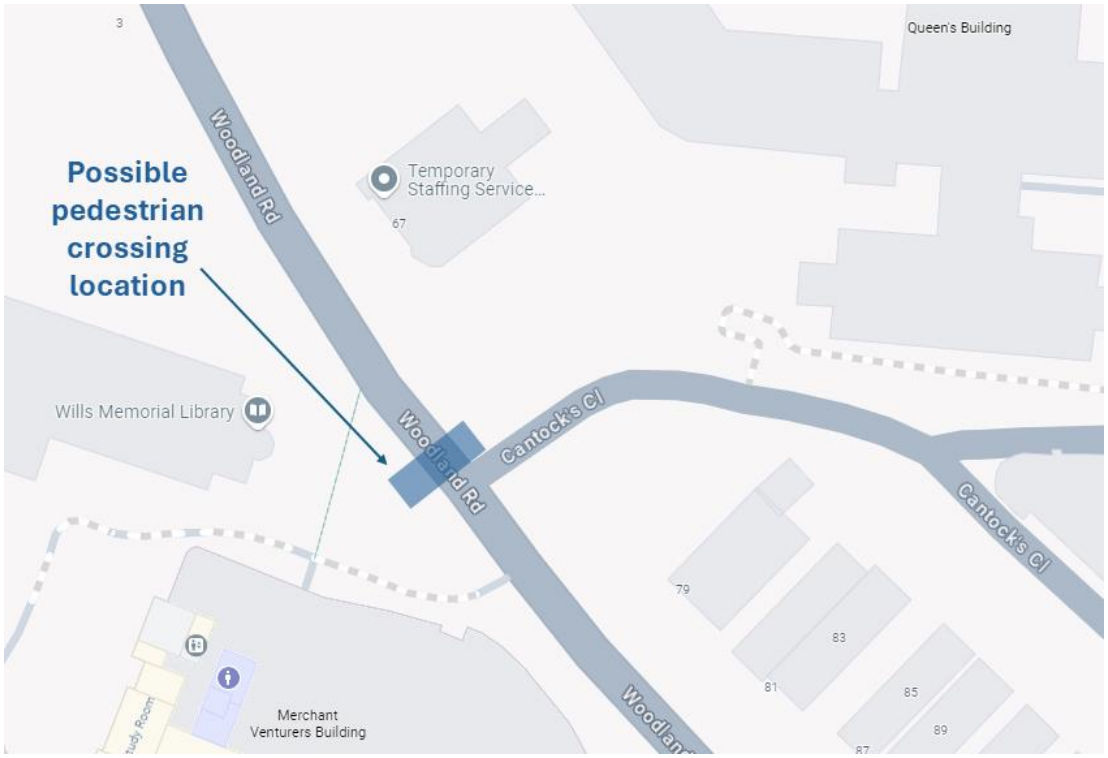


Figure 1: map of proposed pedestrian crossing location within the University precinct.

Traffic from nearby traffic sensors will be made available exclusively to you, the transport consultants tasked with making a recommendation about these plans.

Considerations around this scheme include road safety, smooth traffic flow, and vehicle access to nearby schools.

1. Complete the "PEDESTRIAN CROSSING ASSESSMENT FRAMEWORK" by Leeds City Council (provided on Blackboard). You will have to come up with scores for sections 1-4 in this document. Use the data we provide (section 4; you can also ask for more), online searches (section 2 and 3), and (if applicable) your knowledge of the location or bold guesswork to complete the assessment (section 1).
    a. List the stakeholders, including those directly involved and those indirectly affected by the proposed pedestrian crossing.
    b. Perform a time-series decomposition on vehicle, pedestrian, and cyclist data from the site (worked example in data-driven modelling lab sheet). Is the additional information you get from this useful for assessing the proposed crossing?
    c. Looking at the counts and tracks of road users from the Woodland road sensor (see figure 2 below), as well as considering the sensor location, what issues are there with using this count data for the assessment, and what issues could there be with the positioning and ultimate use of the pedestrian crossing.

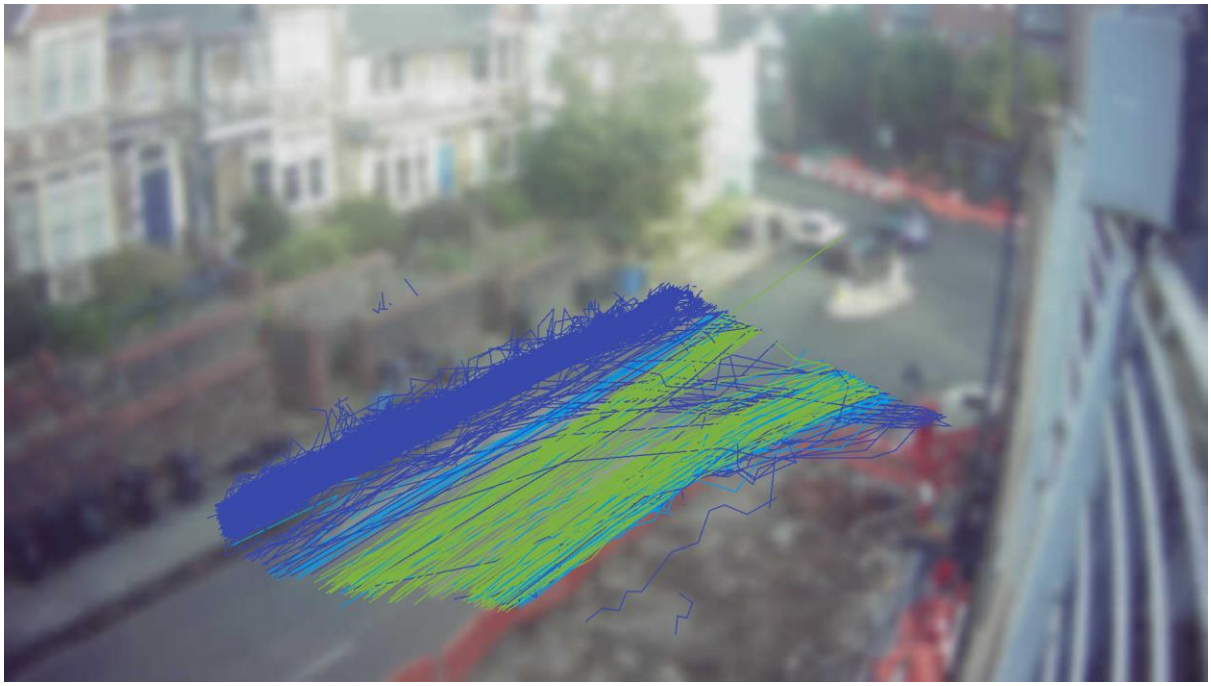


Figure 2: road user tracks from Woodland road Traffic sensor (pointing towards junction with Park Row). Blue: pedestrians, Light blue: cyclists, green: cars. Data is from 15th October 2024, 9.10am-10.10am.

Evidently, the official pedestrian crossing assessment framework we explored above, does not include any transport models. In the remainder of the lab, we will explore how modelling could be used in such an assessment.

2. **Model 1 (data extrapolation).** The problem with the data we provide you with, already initially explored in 1c, is that it does not cover the exact location of the crossing. The assessment framework suggests traffic and pedestrian surveys (these consist of typically – still! – people counting traffic manually), which is expensive. We will now explore how modelling could help with extrapolating, and/or with deciding on what additional data we need.
    a. Separately for cars and pedestrians, draw networks for the in- and out-flows at the proposed pedestrian crossing site. Explain why for car traffic, the data provided may be adequate, whilst for pedestrian traffic it may not be.
    b. Based on your answer to part a, what additional data on pedestrian traffic would you need, and how could it be recorded.
    c. Devise a time-series model that uses the data provided and simulations of the additional data you think you need to obtain an estimate for pedestrian traffic at the crossing site (the simulations of the additional data may also require an appropriate time-series model).
    d. Based on your updated counts and assuming a signalised pedestrian crossing with fixed red/green time intervals, can you provide an estimate for the cost of the crossing to motorists, and its benefit for pedestrians in terms of time delays/savings (ignoring safety aspects).
    e. *Discuss how a direct demand model (for pedestrians) and a version of the classical four-steps developed for Bristol (for cars; see case study on Bristol Clean Air Zone) could be used as alternative routes for predicting traffic at the crossing site. (**Extension: can you program an example for how to do this? Ask if you need more traffic data*).

3. **Model 2 (microsimulation on crossing usage).** In a previous lab (#4) we looked at pedestrian models using the social-force model (SFM) and car-following models using the optimal-velocity model (OVM). Now, in order to study the interaction of pedestrians and vehicles, the SFM and OVM models have been combined into a shared modelling framework (source code available on blackboard as **tmm_lab_06_microsimulation_code.zip** (see Fig. 3):
    a. Run the provided code **intersection_model.py** and experiment with how changing the inflow rate of cars and pedestrians will affect the dynamics of the simulation (the relevant variables are called **cars_inflow** and **peds_inflow**).
    b. Simulate the following three cases: (i) cars_inflow = peds_inflow = 0.1 (s^{-1}); (ii) cars_inflow = peds_inflow = 0.2 (s^{-1}); (iii) cars_inflow = peds_inflow = 0.3 (s^{-1}). For each of the three simulations, calculate the mean journey time for each of the two transport modes and explain why one of the transport modes has a near constant journey time whereas the other transport mode has a varying mean journey time for the three different cases. What are the implications of these results on the viability of a signalled vs non-signalled crossing?
    c. Critically analyse the choices made in the provided model and list any limitations and assumptions that you deem important to highlight in the context of the pedestrian crossing between the Merchant Venturers Building and Queens Building.

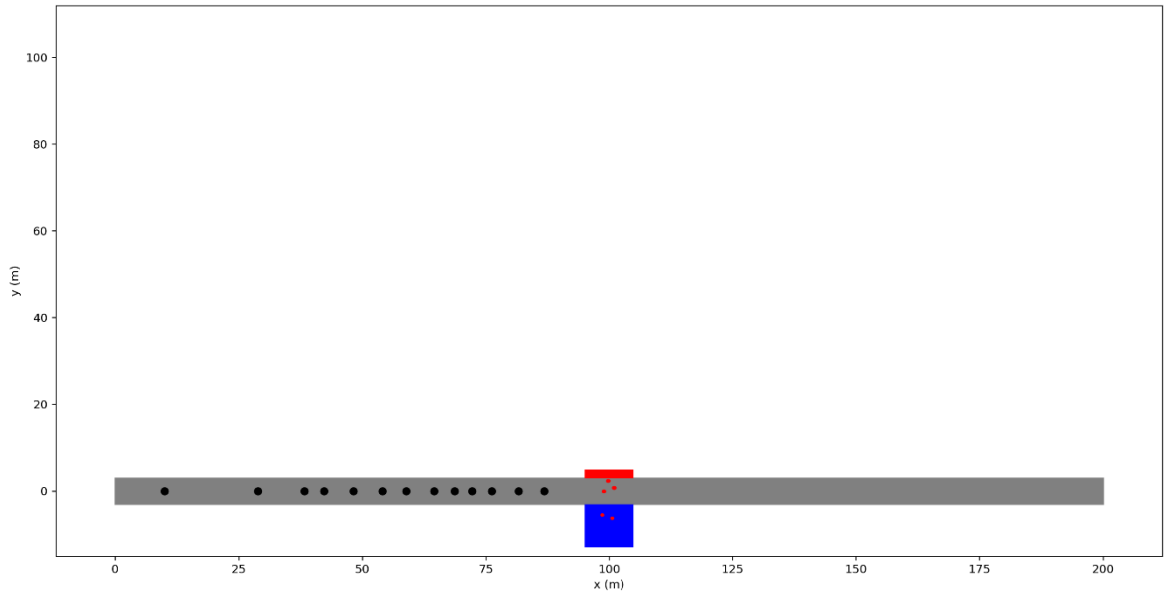


Figure 3: Simulated vehicle-pedestrian intersection where pedestrians (red markers) try to cross a road from the bottom blue area to the top red are at the other side of the road. Python code available for download on Blackboard.

4. **Model 3 (wider effects in the traffic network).** In this question, we will consider the possibility of vehicle congestion caused by the pedestrian crossing, and how this could impact on traffic in the wider area.
    a. One extreme possibility could be that traffic could back up onto Park Row (which is substantially more busy for vehicle traffic). To investigate the possibility of this happening, try either to adapt the traffic simulations on Martin Treiber's website (https://traffic-simulation.de/intersection.html) and/or use the code used in the previous question to study congestion formation and the conditions (incoming flow) under which congestion (queue length) could become an issue. Comparing this to the data for car traffic on Woodland road, do you think this could be an issue?
    b. Using this type of simulation is only an approximation, list reasons for why.
    c. Discuss how building the pedestrian crossing could affect route assignments in the local area.